BEYOND DIAMONDS

A Predictive Analysis of Botswana's Export Diversification

DSS 680 – Predictive Analytics Using SAS Viya

A Foundation Under Strain

Diamonds built Botswana's economy. Lab-grown substitution and weak natural-diamond demand are now testing it.

-5.4%

Q4 2025 GDP growth, year-over-year

driven largely by a mining-sector collapse

-47%

Mining sector contraction, Q4 2025

even as manufacturing & agriculture grew

BBB-

Sovereign credit rating downgrade

S&P cited diamond market weakness directly

Botswana's diamond slowdown makes diversification urgent. This project asks which exports show the clearest evidence of future growth.

The Research Question

Which non-diamond product categories show the strongest and most reliable potential for export growth from 2024 to 2028?

Prediction, not causation

The models forecast export performance; they do not estimate the effect of investment or policy.

Built from real trade history

Product-level UN Comtrade export and import data cover Botswana from 2000 through 2023.

A five-year forecast horizon

Separate models predict 2024–2028, with validation results reported for every horizon.

Data Foundation

Historical features are built first, then models are tested only on later years.

24

years of history

2000–2023, UN Comtrade

98

product categories

UN export and import data for Botswana

1,764

model observations

chronologically partitioned

98

products scored

for 2024–2028 predictions

2000–2004 build the historical variables; 2005–2018 train the models; 2019–2022 test them.

The final 2023 observations generate the forecasts. Category 71 and 99 are excluded only from the final recommendation list.

Eight models tested in SAS Viya

The three required models are shown clearly, with five additional models used for comparison.

Required models

- Linear Regression
- Decision Tree
- Neural Network

Additional models

- Stepwise and LASSO Regression
- Gradient Boosting and Forest
- Support Vector Machine

How models were compared

- 1,372 training observations
- 392 validation observations
- Validation ASE selected the champion
- Each horizon was modeled separately

Results for the three required models

All three were trained and validated on the same chronological data partition.

Linear Regression

Decision Tree

Neural Network

Validation results:

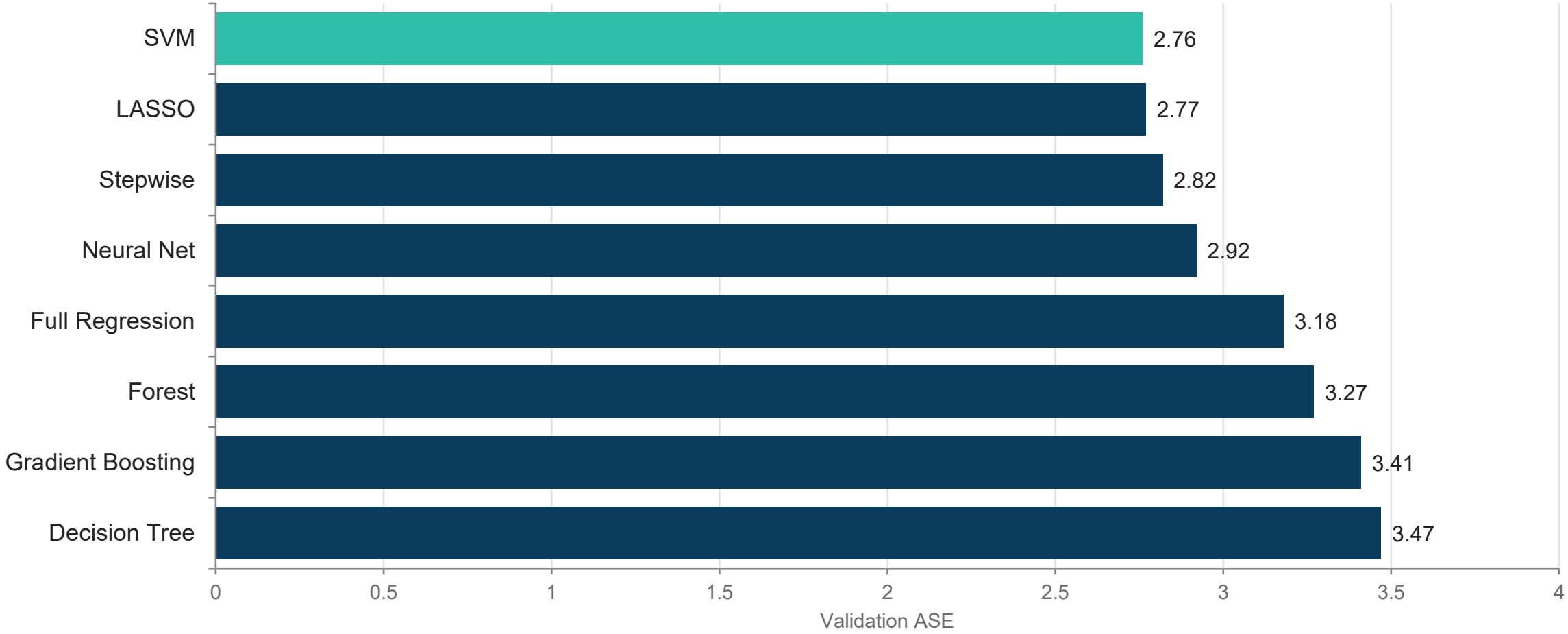
Linear Regression: R^2 0.786, ASE 3.178.

Decision Tree: R^2 0.766, ASE 3.473.

Neural Network: R^2 0.803, ASE 2.920. The two-hidden-layer neural network performed best among the three required models.

SVM produced lowest error amongst all 8 models

Validation Average Squared Error (ASE) was the selection measure; lower values are better.



Linear-kernel SVM selected: ASE 2.758 and validation R^2 0.814

Recent exports carried the strongest signal

The interpretable regression and tree models repeatedly emphasized the same core information.

Current exports

3-year average

Annual imports

What this means:

Products with larger and more consistent recent exports were more likely to maintain stronger future export values. Imports also added useful information about domestic demand and trade activity.

A Critical Check

The model doesn't replace judgment — it still needs a human check.



Ch. 81 — Base Metals — ranked #1 by predicted growth (+87.8%)

- 2022 export value spiked to $2.99M — nearly 15× the surrounding years
- 2023 crashed back to $309K, a 90% reversal
- 5-year volatility score: among the highest of any product in the dataset
- The same volatility measure showed the strongest overall correlation with the target — negatively

Decision: excluded from the core recommendation, despite ranking highest on the raw metric — a single anomalous year, not sustained momentum.

Leading 2024 export-growth candidates

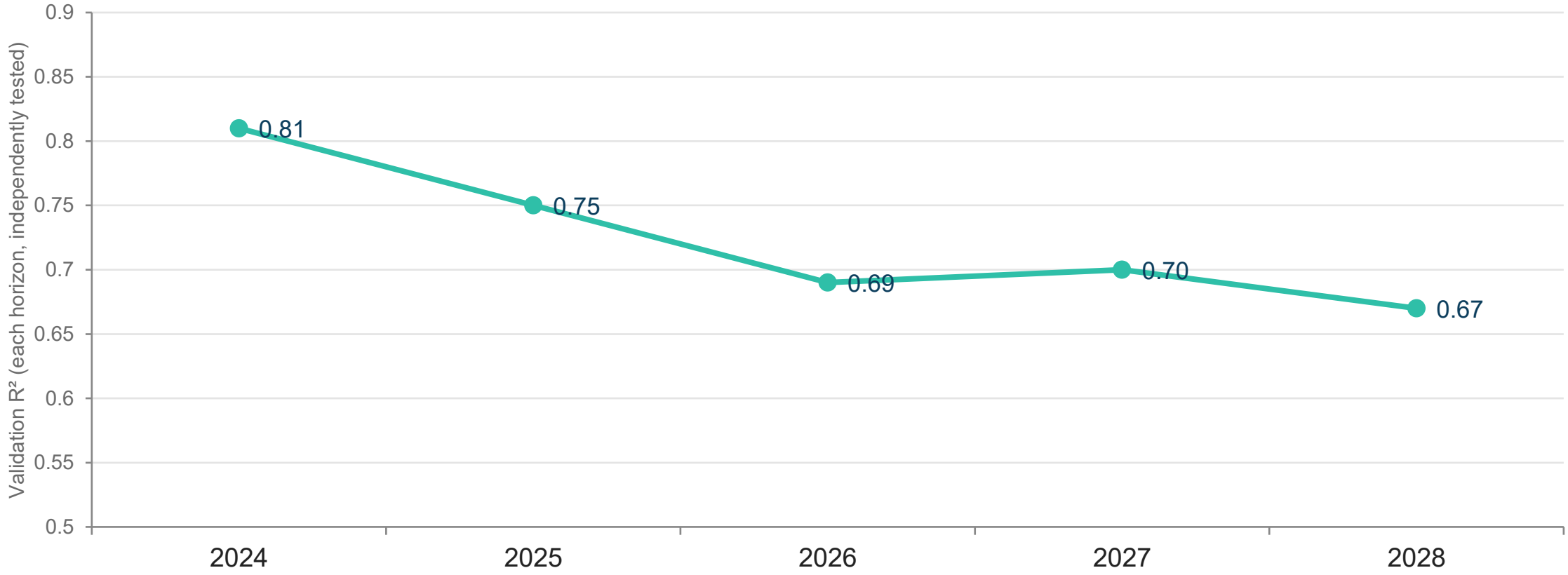
Highest SVM-predicted growth among products with at least $100,000 in 2023 exports

Product	2023 Exports	Predicted 2024	Growth
Miscellaneous manufactured articles	$107,116	$196,391	+60.6%
Cereals	$839,478	$1,527,147	+59.8%
Sugars and sugar confectionery	$332,216	$561,834	+52.5%
Ceramic products	$125,255	$173,291	+32.5%
Mineral fuels & oils	$36,738,787	$49,090,812	+29.0%
Beverages, spirits and vinegar	$1,956,453	$2,542,725	+26.2%
Pulp of wood / cellulosic material	$879,960	$1,127,243	+24.8%
Printed books & publications	$110,557	$139,150	+23.0%

Chapter 81 (Metals) is omitted because its apparent growth was driven by an unstable one-year spike.

Forecast reliability declines over time

Validation R^2 remains useful across the horizon, but later forecasts should be read more cautiously.



A separate model was trained and validated for each year from 2024 through 2028.

Limitations & Future Work

Export growth is not GDP growth

The forecasts rank export opportunities; they do not estimate each product's contribution to national GDP.

No causal investment estimate

The results cannot show how much additional export growth a specific investment or policy would create.

Later forecasts are less certain

Validation R^2 falls from 0.81 for 2024 to 0.67 for 2028, so later results are directional.

Trade data end in 2023

At collection, UN Comtrade had Botswana data through 2023. This made 2024–2028 genuine future forecasts.

A portfolio offers the strongest path beyond diamonds

SVM performed best overall. Cereals, sugars, ceramics, beverages, and several other products emerged as credible growth candidates for further investigation, not automatic investment decisions.

Thank you

DSS 680 Final Project — Seboni
This presentation was made with the assistance of Claude.